

**MESS FOOD WASTAGE TRACKER AND
MANAGEMENT SYSTEM**

Software Requirements Specification (SRS) Report

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Chapter 1

Abstract

Food wastage in institutional and college mess facilities is a common operational problem. A considerable amount of food may be prepared without accurate information about the expected number of students, while unexpected absences, changes in demand, and inaccurate estimations may further increase the quantity of leftover food.

The proposed **Mess Food Wastage Tracker and Management System** is a software system designed to record, monitor, analyze, and manage food wastage in a college or institutional mess. The system will allow mess administrators to maintain information about meals, the number of students expected to eat, the quantity of food prepared, the quantity consumed, and the quantity remaining or wasted.

The system will provide useful information through reports and summaries so that mess administrators can identify patterns of wastage and make better decisions regarding food preparation. Students may also be allowed to indicate their meal attendance in advance, helping the mess management estimate the expected demand more accurately.

The system is proposed to follow the **Agile Software Development Model**. Agile development is suitable because the system can be developed incrementally and improved through continuous feedback from students, mess staff, and administrators.

Chapter 2

Objective

The primary objective of the Mess Food Wastage Tracker and Management System is to reduce unnecessary food wastage by providing a systematic method for recording and analyzing meal-related information.

The major objectives of the proposed system are:

1. To record the number of students expected to attend each meal.
2. To maintain records of food prepared for breakfast, lunch, snacks, and dinner.
3. To record the amount of food consumed and the amount of food left over.
4. To calculate and track food wastage for individual meals and over longer periods.
5. To identify meals, days, or food items with consistently high wastage.
6. To help mess administrators make better food preparation decisions.
7. To provide reports and summaries for management and analysis.
8. To provide a simple interface for students and mess staff.
9. To maintain historical records for future planning and comparison.
10. To encourage efficient utilization of food resources.

Chapter 3

Introduction

3.1 Purpose

The purpose of this Software Requirements Specification is to define the requirements of the Mess Food Wastage Tracker and Management System. The document describes the functionality, users, constraints, interfaces, and quality requirements of the proposed system.

The SRS provides a common understanding between developers, users, administrators, and other stakeholders regarding what the system is expected to accomplish.

3.2 Scope

The system is intended for use in a college, university, hostel, institutional, or similar mess environment.

The system will maintain information related to:

- Students and users.
- Daily meal schedules.
- Expected meal attendance.
- Food preparation quantities.
- Food consumption quantities.
- Leftover and wasted food.
- Food items and menus.

- Historical wastage data.
- Reports and statistical summaries.

The system will primarily focus on tracking and management of food wastage. It is not intended to replace complete hostel management, accounting, procurement, or kitchen inventory systems, although future versions may integrate with such systems.

3.3 Definitions

Term	Definition
Student	A person who is eligible to use the institutional mess.
Mess Administrator	Person responsible for managing the mess system and monitoring reports.
Mess Staff	Personnel responsible for food preparation and recording meal information.
Meal	A scheduled food service such as breakfast, lunch, snacks, or dinner.
Expected Attendance	Estimated number of students expected to consume a particular meal.
Prepared Quantity	Quantity of food prepared for a particular meal.
Consumed Quantity	Quantity of food actually consumed by users.
Wasted Quantity	Quantity of prepared food that remains unused or is discarded.
Wastage Rate	Percentage of prepared food that is wasted.
Menu	List of food items planned for a particular meal.
Report	A summarized presentation of recorded food and wastage data.

3.4 Overview

The system will consist of a user-facing component through which students can indicate meal attendance and a management component through which authorized staff can enter meal and wastage information.

The system will collect data throughout the day and use the collected information to generate useful summaries. For example, if a particular meal repeatedly has a high amount of

leftover food, the administrator can review previous attendance and preparation records before adjusting future preparation quantities.

Chapter 4

SDLC Model

4.1 Selected SDLC Model

The **Agile Software Development Model** has been selected for the development of the Mess Food Wastage Tracker and Management System.

Agile is an iterative and incremental development approach in which software is developed in relatively small increments. Requirements are continuously reviewed and changes can be incorporated during subsequent development cycles.

For the proposed system, the initial version may contain basic features such as student meal attendance and wastage recording. Additional functionality, such as analytics, notifications, dashboards, or prediction of meal demand, can be introduced in later iterations.

4.2 Advantages of the Agile Model

The major advantages of using Agile for the proposed system are:

1. **Flexibility:** Requirements can be modified as feedback is obtained from students and mess staff.
2. **Incremental Development:** The system can be developed feature by feature instead of waiting for the complete system before testing.
3. **Continuous Feedback:** Users can review implemented features and provide suggestions for improvement.
4. **Early Delivery:** A basic working system can be delivered early and expanded over time.

5. **Early Detection of Problems:** Errors and usability problems can be identified during individual development iterations.
6. **Suitable for Evolving Requirements:** Actual mess operations may reveal new requirements after the system is introduced, making an adaptable development model useful.

4.3 Disadvantages of the Agile Model

The major disadvantages are:

1. **Requires Continuous User Involvement:** Meaningful feedback from students and mess staff is important for successful development.
2. **Changing Requirements:** Frequent changes may make planning and estimation more difficult.
3. **Documentation May Be Reduced:** If not properly managed, emphasis on rapid iterations can result in insufficient documentation.
4. **Scope Creep:** New feature requests may continuously increase the scope of the project.
5. **Requires Good Team Coordination:** Frequent communication and coordination are necessary among team members.

4.4 Reason for Selecting Agile

Agile has been selected because the exact requirements of a mess food management system may not be completely known at the beginning of the project.

Different users may have different expectations. Students may want a simple way to indicate whether they will attend a meal, while mess staff may require a fast interface for entering prepared and wasted quantities. Administrators may require reports, charts, and historical comparisons.

Therefore, developing the system incrementally allows the team to implement the most important features first and improve the system based on stakeholder feedback.

The proposed Agile development process can be organized into iterations such as:

1. Requirements and initial planning.

2. User and meal management.
3. Meal attendance and demand recording.
4. Food preparation and wastage recording.
5. Reports and analytics.
6. Testing, feedback, and refinement.

4.5 SDLC Model Diagram

Insert the Agile SDLC diagram here.

Chapter 5

Overall Description

5.1 Product Perspective

The Mess Food Wastage Tracker and Management System will function as an independent information management system for a college or institutional mess.

The system will receive input from students and authorized mess staff. The collected information will be stored in a central database and processed to generate summaries and reports.

The system may be implemented as a web-based application so that students and administrators can access it from computers or mobile devices connected to the appropriate network.

5.2 Product Functions

The main functions of the system are:

1. User registration and authentication.
2. Role-based access control.
3. Student meal attendance marking.
4. Daily menu management.
5. Expected meal count calculation.
6. Food preparation quantity recording.
7. Food consumption quantity recording.

8. Food wastage quantity recording.
9. Wastage percentage calculation.
10. Daily, weekly, and monthly reports.
11. Identification of high-wastage meals.
12. Historical data storage.
13. Administrative management of users and system data.

5.3 User Characteristics

The proposed system will mainly have three types of users.

5.3.1 Student

Students will use the system to view meal information and indicate whether they intend to attend a particular meal. Students will have limited access to management functions.

5.3.2 Mess Staff

Mess staff will record information related to food preparation, consumption, leftovers, and wastage. They will require a simple interface because the information may need to be entered during busy kitchen operations.

5.3.3 Administrator

The administrator will have the highest level of access. The administrator will manage users, meals, menus, reports, and system configuration.

5.4 Operating Environment

The proposed system may operate in the following environment:

- Client devices: desktop computers, laptops, tablets, or smartphones.
- Server: local or cloud-based application server.

- Database: relational database management system.
- Network: institutional network or Internet connection.
- Operating systems: Windows, Linux, Android, or other systems supporting a modern web browser.
- Browser: Chrome, Firefox, Edge, Safari, or equivalent modern browser.

5.5 Constraints

The following constraints may affect the system:

1. Accurate reports depend on the accuracy of entered data.
2. Students must have access to the system to record meal attendance.
3. Internet or network availability may be required for a web-based implementation.
4. Mess staff may have limited time for entering data.
5. The system must protect student information from unauthorized access.
6. The system should remain simple enough for non-technical users.

5.6 Assumptions and Dependencies

The system assumes that:

1. Students provide reasonably accurate meal attendance information.
2. Mess staff record prepared and wasted quantities regularly.
3. Authorized users have valid accounts.
4. The database remains available when information is entered.
5. The mess follows a defined meal schedule.

The system may depend on a database server, application server, authentication mechanism, and network connectivity.

Chapter 6

Requirement Specification

6.1 Functional Requirements

The functional requirements describe the operations that the system must perform.

6.1.1 User Authentication and Authorization

- FR1. The system shall allow users to log in using valid credentials.
- FR1. The system shall identify users according to their role.
- FR1. The system shall restrict management functionality to authorized users.
- FR1. The system shall allow administrators to activate or deactivate user accounts.

6.1.2 Student Meal Attendance

- FR2. The system shall display the meals available for a particular day.
- FR2. The system shall allow a student to indicate attendance for a meal.
- FR2. The system shall allow students to modify their attendance before the configured deadline.
- FR2. The system shall record the date, meal, and student associated with the attendance entry.
- FR2. The system shall calculate the expected number of students for each meal.

6.1.3 Menu Management

FR3. The system shall allow authorized users to create a menu for a meal.

FR3. The system shall allow authorized users to modify menu information.

FR3. The system shall store menu history for previous meals.

6.1.4 Food Preparation Recording

FR4. The system shall allow mess staff to record the quantity of food prepared.

FR4. The system shall associate prepared quantities with a date, meal, and food item.

FR4. The system shall allow corrections to be made by authorized users.

6.1.5 Food Consumption and Wastage Recording

FR5. The system shall allow mess staff to record the consumed quantity.

FR5. The system shall allow mess staff to record leftover or wasted quantity.

FR5. The system shall calculate wastage based on recorded food quantities.

FR5. The system shall maintain historical wastage records.

6.1.6 Wastage Analysis

FR6. The system shall calculate the wastage percentage for a meal.

FR6. The system shall allow users with appropriate permissions to view daily wastage.

FR6. The system shall provide weekly and monthly summaries.

FR6. The system shall identify meals with unusually high wastage.

FR6. The system shall support comparison of wastage across different dates and meals.

6.1.7 Report Generation

FR7. The system shall generate daily wastage reports.

FR7. The system shall generate weekly and monthly reports.

FR7. The system shall provide attendance and preparation summaries.

FR7. The system shall provide reports that can assist management in planning future meals.

6.1.8 Administration

FR8. The administrator shall be able to manage student and staff accounts.

FR8. The administrator shall be able to configure meal schedules.

FR8. The administrator shall be able to manage food items and menus.

FR8. The administrator shall be able to view system reports.

6.2 Non-Functional Requirements

The non-functional requirements define the quality attributes and constraints of the system.

6.2.1 Performance Requirements

NFR1. The system should respond to normal user requests within a reasonable time.

NFR1. Frequently used operations such as viewing meals or submitting attendance should not require unnecessary processing.

NFR1. Report generation should complete within an acceptable time for the expected amount of stored data.

6.2.2 Security Requirements

NFR2. The system shall prevent unauthorized access to protected functionality.

NFR2. Passwords shall not be stored in plain text.

NFR2. Users shall only access information permitted by their roles.

NFR2. Administrative operations shall be restricted to authorized personnel.

NFR2. Sensitive user information shall be protected from unauthorized disclosure.

6.2.3 Usability Requirements

- NFR3. The user interface should be simple and understandable.
- NFR3. Common operations should require a minimum number of steps.
- NFR3. The interface should be usable by users with basic computer or smartphone knowledge.
- NFR3. The system should provide understandable validation and error messages.

6.2.4 Reliability Requirements

- NFR4. The system should maintain consistent records during normal operation.
- NFR4. Data should not be lost when a valid transaction has been completed.
- NFR4. The system should provide suitable error handling when an operation cannot be completed.
- NFR4. Regular database backup should be supported.

6.2.5 Scalability Requirements

- NFR5. The system should support an increasing number of students and historical records.
- NFR5. The architecture should permit additional mess facilities or branches to be supported in the future.
- NFR5. Additional reporting and analytical features should be addable without redesigning the complete system.

6.2.6 Maintainability Requirements

- NFR6. The system should use a modular architecture.
- NFR6. Major components should be sufficiently documented.
- NFR6. Changes to one feature should have minimal impact on unrelated features.
- NFR6. The database and application code should be organized so that future enhancements are possible.

6.2.7 Availability Requirements

NFR7. The system should be available during normal mess operating hours.

NFR7. Planned maintenance should preferably occur outside critical meal periods.

Chapter 7

Decision Analysis

7.1 Decision Tree

The decision tree for the system can be based on the decision of whether food preparation should be increased, maintained, or reduced for a particular meal.

The major factors considered may include:

- Expected attendance.
- Historical attendance.
- Quantity of food prepared.
- Previous wastage.
- Current menu.
- Special events or unusual circumstances.

For example, if expected attendance is high and previous wastage is low, the recommended action may be to maintain or increase preparation. If expected attendance is low and previous wastage is consistently high, preparation may be reduced.

Insert the Decision Tree diagram here.

7.2 Decision Table

An example decision table for meal preparation is given below.

Table 7.1: Decision Table for Food Preparation

Condition / Action	Case 1	Case 2	Case 3	Case 4
Expected attendance is high	Y	Y	N	N
Historical wastage is high	N	Y	N	Y
Recommended preparation	Increase	Maintain/Review	Maintain	Reduce

The exact thresholds for “high attendance” and “high wastage” can be configured by the system administrator.

Chapter 8

System Design

8.1 DFD Level 0

The Level 0 Data Flow Diagram will represent the system as a single major process and show interaction with external entities such as students, mess staff, and administrators.

Insert DFD Level 0 diagram here.

8.2 DFD Level 1

The Level 1 DFD will decompose the main system into major processes such as:

1. User and Authentication Management.
2. Meal and Menu Management.
3. Attendance Management.
4. Food Preparation Management.
5. Wastage Management.
6. Report Generation.

Insert DFD Level 1 diagram here.

8.3 Structure Chart

The structure chart will represent the hierarchical organization of the software modules.

The proposed main modules are:

- Authentication Module
- Student Module
- Meal Management Module
- Attendance Module
- Food Recording Module
- Wastage Analysis Module
- Reporting Module
- Administration Module

Insert Structure Chart here.

8.4 Use Case Diagram

The principal actors in the system are:

- Student
- Mess Staff
- Administrator

The major use cases include:

- Login
- View Menu
- Mark Meal Attendance
- Modify Meal Attendance
- Record Food Prepared
- Record Food Consumed
- Record Food Wasted

- View Wastage
- Generate Reports
- Manage Users
- Manage Meals and Menus

Insert Use Case Diagram here.

8.5 State Chart Diagram

The state chart can represent the lifecycle of a meal record.

A typical sequence of states can be:

1. Meal Scheduled
2. Attendance Open
3. Attendance Closed
4. Food Preparation
5. Meal In Progress
6. Meal Completed
7. Wastage Recorded
8. Report Generated

A meal may transition from one state to another depending on time, administrative actions, and completion of the meal.

Insert State Chart Diagram here.

8.6 Class Diagram

The major classes expected in the proposed system are:

- User
- Student

- MessStaff
- Administrator
- Meal
- Menu
- FoodItem
- Attendance
- FoodRecord
- WastageRecord
- Report

Possible relationships include:

- A Student records Attendance for a Meal.
- A Meal has a Menu.
- A Menu contains multiple FoodItems.
- A Meal has FoodRecord entries.
- A FoodRecord may result in a WastageRecord.
- Reports are generated from stored meal and wastage information.
- Administrators manage users and configuration data.

Insert Class Diagram here.

Chapter 9

Behavioral Design

9.1 Activity Network Diagram

The activity network represents the major activities involved in managing food wastage.

A representative workflow is:

1. Create or retrieve the daily meal schedule.
2. Display meal information to students.
3. Collect expected meal attendance.
4. Calculate expected attendance.
5. Prepare food according to the estimated requirement.
6. Conduct the meal.
7. Record consumed quantity.
8. Record leftover or wasted quantity.
9. Calculate wastage.
10. Store the result.
11. Generate reports and summaries.

Several activities can be performed in parallel. For example, administrative preparation of menu information and student attendance collection may occur during overlapping time periods.

Insert Activity Network Diagram here.

9.2 Interaction Diagram

The interaction diagram will show communication between the main objects involved in a selected operation.

One suitable scenario is **Recording Meal Wastage**.

The interaction can be described as follows:

1. Mess staff selects the completed meal.
2. The user interface sends the selected meal information to the food management component.
3. The food management component retrieves the corresponding food records.
4. Mess staff enters the consumed and wasted quantities.
5. The system validates the entered quantities.
6. The system calculates the wastage value and percentage.
7. The system stores the wastage record in the database.
8. The system confirms successful recording to the user.

Insert Interaction/Communication Diagram here.

9.3 Sequence Diagram

A suitable sequence diagram can represent the process of a student marking attendance for a meal.

The sequence can be described as:

1. Student logs into the system.
2. System authenticates the student.
3. Student requests the available meals.
4. System retrieves the meal schedule.
5. Student selects a meal.

6. Student submits attendance.
7. System validates the request.
8. System stores the attendance record.
9. System updates the expected attendance count.
10. System displays confirmation.

An alternative sequence diagram may represent the process of recording food wastage by mess staff.

Insert Sequence Diagram here.

Chapter 10

Additional Requirement Details

10.1 Input Requirements

The system may receive the following inputs:

- User login credentials.
- Student meal attendance.
- Meal schedules.
- Menu and food item information.
- Prepared food quantity.
- Consumed food quantity.
- Wasted or leftover quantity.
- Administrative configuration information.

10.2 Output Requirements

The system will provide outputs such as:

- Daily meal attendance.
- Expected meal count.
- Food preparation records.

- Food wastage quantity.
- Wastage percentage.
- Daily wastage summary.
- Weekly and monthly reports.
- High-wastage meal identification.
- Historical comparisons.

10.3 Data Requirements

The database should maintain information regarding:

1. Users.
2. User roles.
3. Meals.
4. Menus.
5. Food items.
6. Attendance.
7. Food preparation.
8. Food consumption.
9. Wastage.
10. Reports and historical records.

Chapter 11

Conclusion and Future Work

11.1 Conclusion

The Mess Food Wastage Tracker and Management System provides a systematic approach to monitoring and managing food wastage in an institutional mess.

The proposed system records expected attendance, food preparation, consumption, and wastage information. By maintaining historical records and generating reports, the system can help mess administrators understand wastage patterns and make more informed decisions about food preparation.

The Agile SDLC model has been selected because it allows the system to be developed incrementally and refined according to feedback from students, mess staff, and administrators.

The system is intended to improve operational awareness and reduce unnecessary food wastage while providing an organized record of meal-related data.

11.2 Future Work

Several improvements can be incorporated into future versions of the system.

1. **Food Demand Prediction:** Machine learning techniques can be used to predict expected meal demand from historical attendance and wastage data.
2. **Mobile Application:** A dedicated Android or cross-platform mobile application can make meal attendance easier for students.
3. **Notifications:** The system can send reminders to students who have not submitted meal attendance before a deadline.

4. **Dashboard and Visualization:** Interactive charts can be added to show trends in food wastage.
5. **Cost Analysis:** The system can estimate the monetary value associated with wasted food.
6. **Inventory Integration:** The system can be integrated with inventory and procurement systems.
7. **Multiple Mess Support:** A single platform can support multiple mess facilities.
8. **Automated Measurement:** Sensors or weighing systems could be integrated to reduce manual entry of food quantities.
9. **Sustainability Metrics:** Future versions can provide additional sustainability indicators related to food waste.

Chapter 12

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